

An Empirical Study of Algorithmic Stability and Regularization in Linear Models

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Abstract

This study empirically investigates the framework of algorithmic stability proposed by Bousquet and Elisseeff, exploring its direct mathematical link to generalization performance in linear regression models. We systematically evaluate unregularized Ordinary Least Squares (OLS), Ridge (L2), and Lasso (L1: [3]) regression across a logarithmic spectrum of regularization strengths using both highly controlled synthetic environments and real-world data (the Scikit-Learn Diabetes dataset). Algorithmic stability is quantified empirically using a Leave-One-Out (LOO) methodology to measure the average variance in model predictions. Our experimental results confirm that appropriate regularization significantly reduces algorithmic instability, successfully mitigating overfitting and producing the characteristic U-shaped curve in the generalization error. Furthermore, extended analyses demonstrate that the intrinsic necessity for regularization decreases as dataset size increases, due to the natural stabilizing effect of the Law of Large Numbers. Finally, a geometric comparison between L1 and L2 penalties provides empirical proof of Lasso’s capability to enforce model sparsity and active feature selection. Ultimately, this report validates algorithmic stability not merely as a theoretical concept, but as a robust, operational proxy for model optimization and generalization.

1 Introduction

Supervised machine learning fundamentally relies on Empirical Risk Minimization (ERM), a framework where models are trained to minimize a predefined loss function over a given set of training data. The ultimate goal, however, is not merely to memorize the training set, but to achieve high generalization performance on unseen data. Classical statistical learning theory explains this generalization capability through the lens of hypothesis complexity—using tools such as uniform convergence and Vapnik-Chervonenkis (VC) dimension. Within this classical framework, regularization is introduced as a mathematical penalty to restrict the hypothesis space, thereby controlling the model’s complexity and mitigating the risk of overfitting. While classical complexity measures provide a robust foundational theory, they occasionally fail to fully explain why certain algorithms generalize exceptionally well, even in high-dimensional settings where traditional complexity bounds become loose or uninformative. To address this theoretical gap, Olivier Bousquet and André Elisseeff proposed here [1] a paradigm shift: evaluating generalization through the lens of Algorithmic Stability. Instead of focusing exclusively on the complexity of the chosen hypothesis class, stability theory investigates the inherent sensitivity of the learning algorithm itself to minor perturbations in the training dataset. Intuitively, an algorithm is considered stable if a small change in the input data—such as the removal or replacement of a single training sample—does not cause a disproportionate shift in the resulting predictive model. Formally, this concept is often measured using the Leave-One-Out (LOO) methodology. Let S be a training set of size N , and let $S^{\setminus i}$ denote the same dataset with the i -th instance removed. An algorithm demonstrates high algorithmic stability if the difference in predictions (or loss) between

the model trained on S and the model trained on $S^{\setminus i}$ remains consistently bounded across all points. According to Bousquet and Elisseeff, this stability is not merely a desirable algorithmic property, but a direct mathematical proxy for good generalization.

Objective of the Report

This report aims to empirically validate the theoretical framework of algorithmic stability. By bridging the gap between theoretical bounds and practical implementation, the study investigates the intricate relationship between regularization strength (λ), algorithmic stability, and generalization error. Specifically, the experiments compare Unregularized Ordinary Least Squares (OLS) with Ridge Regression (L2 regularization: [2]) across a spectrum of penalty values. Through the evaluation of synthetic and real-world datasets, this report will provide empirical evidence demonstrating that properly tuned regularization inherently increases an algorithm’s stability, which in turn leads to a minimization of the test error, effectively navigating the classic bias-variance trade-off. Furthermore, the report extends this analysis to explore the mitigating effect of dataset size on model instability, as well as the structural differences in stability and sparsity introduced by Lasso (L1) regularization.

2 Methodology and Experimental Setup

To empirically investigate the relationship between stability, regularization, and generalization, a controlled experimental pipeline was developed from scratch. This section outlines the regression models, the optimization algorithms, the evaluation metrics, and the data generation process used in the study.

2.1 Regression Models and Regularization Techniques

The core of the experiment relies on training linear regression models under different regularization regimes. Given a dataset with N samples and D features, the objective is to find a weight vector w that minimizes a specific loss function. Unregularized Ordinary Least Squares (OLS): Serves as the baseline model. It minimizes the residual sum of squares without any penalty on the weights. In high-dimensional regimes ($N \approx D$), OLS is highly susceptible to fitting the background noise, leading to severe overfitting. Ridge Regression (L2 Penalty): Introduces a penalty proportional to the squared magnitude of the weights. The objective function becomes:

$$L_{Ridge}(w) = \text{MSE} + \lambda \sum_{j=1}^D w_j^2$$

where $\lambda \geq 0$ controls the regularization strength. The model was implemented using a closed-form solution. For numerical stability, rather than explicitly computing the matrix inverse, the weights were found by solving the linear system $(X^T X + \lambda I)w = X^T y$. Note that the bias term (intercept) was explicitly excluded from the regularization penalty to prevent arbitrary shifting of the decision boundary. Lasso Regression (L1 Penalty): Used for the extended comparative analysis, Lasso applies a penalty proportional to the absolute value of the weights:

$$L_{Lasso}(w) = \text{MSE} + \lambda \sum_{j=1}^D |w_j|$$

Unlike Ridge, the L1 penalty geometry features sharp corners, meaning its derivative is undefined at zero and it lacks a closed-form solution. To train this model, an iterative Coordinate Descent algorithm was implemented from scratch. The algorithm updates one weight at a time using the Soft Thresholding operator, which actively forces less important feature weights to exactly zero, thereby inducing model sparsity (feature selection).

2.2 Evaluation Metrics

To quantify the theoretical concepts proposed by Bousquet and Elisseeff, the models were evaluated across three distinct dimensions:

- **Generalization Performance:** Measured via the Mean Squared Error (MSE) on a fixed, strictly unseen test set.
- **Algorithmic Stability:** Estimated empirically using a Leave-One-Out (LOO) approach. For a training set of size N , the model is retrained N times, each time systematically removing the i -th training instance. Stability is quantified as the average absolute change in predictions across the fixed test set:

$$\text{Instability} = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{M} \sum_{j=1}^M \left| \hat{y}(x_j) - \hat{y}^{(i)}(x_j) \right| \right)$$

where M is the size of the test set, $\hat{y}$ are the base model predictions, and $\hat{y}^{(i)}$ are the predictions of the LOO model. Note: Higher values of this metric indicate high algorithmic instability.

- **Model Sparsity:** Specifically for the L1 vs L2 comparison, sparsity was measured by counting the exact number of weights reduced to 0.0 at each λ step.

2.3 Dataset Generation and Preprocessing

To strictly control the signal-to-noise ratio and guarantee a baseline state of overfitting, a synthetic dataset was generated. The input features X were sampled from a standard multivariate Gaussian distribution. A ground truth weight vector w_{true} was uniformly sampled, and the target variable y was defined as a deterministic linear function of the inputs, corrupted by additive Gaussian noise to simulate real-world measurement errors:

$$y = Xw_{true} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

For the primary experiments, the dataset was deliberately configured with a low sample-to-feature ratio (e.g., $N = 150$, $D = 15$ or $N = 60$, $D = 50$) paired with high noise ($\sigma = 25$). This configuration ensures that the unregularized OLS model severely overfits the training data, providing a clear starting point to observe the corrective effects of λ . Prior to training, the datasets underwent strict preprocessing: input features were standardized (zero mean, unit variance), and the target variable was centered. To prevent any data leakage, scaling parameters were computed exclusively on the training split and subsequently applied to the test split.

3 Results: Regularization, Stability, and Generalization

To evaluate the primary hypothesis formulated by Bousquet and Elisseeff—that algorithmic stability is intrinsically linked to generalization performance—the unregularized Ordinary Least Squares (OLS) model was compared against Ridge Regression across a logarithmic spectrum of regularization strengths (λ). The primary experiment was conducted on the synthetic dataset ($N = 150$, $D = 15$) with significant additive noise to ensure an initial state of overfitting.

Observing the dynamics of the two curves reveals a textbook demonstration of the bias-variance trade-off, directly modulated by the stability of the algorithm.

The Baseline: Unregularized OLS ($\lambda \approx 0$)

At the extreme left of the plot, the unregularized OLS model exhibits the highest instability alongside a disproportionately high test error. Because the model is entirely unconstrained, it attempts to perfectly fit every data point, including the synthetic noise. Consequently, the removal of a single instance during the Leave-One-Out validation causes violent fluctuations in the learned decision boundary. The model is highly flexible but entirely unstable, leading to poor generalization.

The "Sweet Spot": Optimal Ridge Regression

As the regularization strength λ increases, the red curve demonstrates a monotonic exponential decrease in algorithmic instability. By penalizing large weights, the algorithm becomes less sensitive to individual data points. Crucially, this gain in stability immediately translates to a dramatic reduction

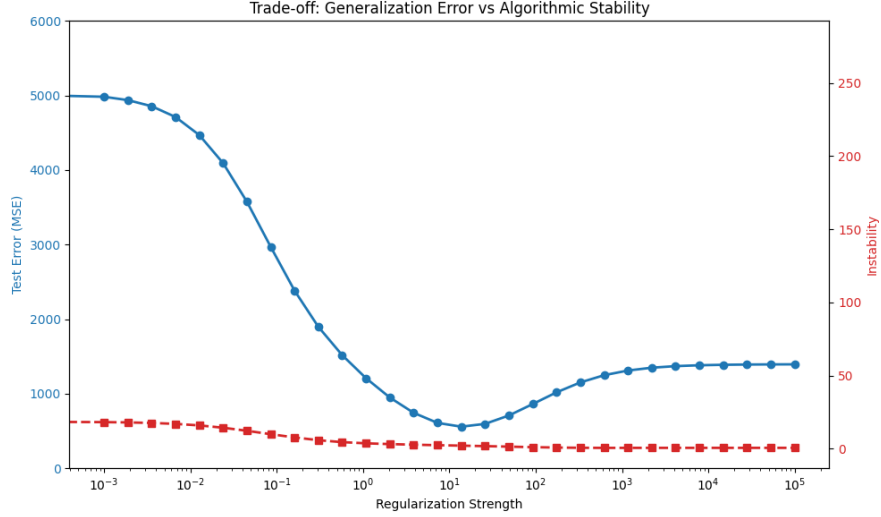


Figure 1: Dual impact of the L2 penalty on both the Test Error (MSE) and the Algorithmic Instability

in the generalization error (the blue curve). The Test Error plummets, forming a distinct "U-shape" curve, and reaches its absolute minimum at an intermediate value of λ . At this optimal point (the "Best Ridge"), the model achieves the perfect balance: it is stable enough to ignore the background noise, but flexible enough to capture the underlying linear trend of the data.

Extreme Regularization and Underfitting

To further dissect this dynamic, a discrete comparative analysis was extracted from the continuum, highlighting four distinct stages of the regularization process.

	Test error	Instability
OLS ($\lambda = 0$)	32811.49	278.46
Med Ridge ($\lambda = 0.09$)	2968.87	9.99
Best Ridge ($\lambda = 13.74$)	557.07	2.17
Worst Ridge ($\lambda = 100000.00$)	1393.53	0.67

Table 1: λ effects on regularization

As summarized in Table 1, pushing λ towards extreme values ($\lambda = 10^5$) introduces a critical phenomenon. While the algorithmic instability drops to its absolute minimum (0.67), making the "Worst Ridge" the most stable algorithm tested, the Test Error paradoxically increases and plateaus. This occurs because the excessive penalty forces all feature weights aggressively toward zero. The model degrades into a flat horizontal line predicting the global mean of the target variable. The LOO variation is negligible because removing a point barely affects a global mean, but the model has entirely lost its predictive power (Underfitting). These empirical results firmly validate the theoretical framework: while a lack of stability causes overfitting, absolute stability without flexibility causes underfitting. Optimal generalization is achieved precisely at the inflection point where regularization provides sufficient algorithmic stability without inducing severe bias.

4 Impact of Dataset Size on Algorithmic Stability

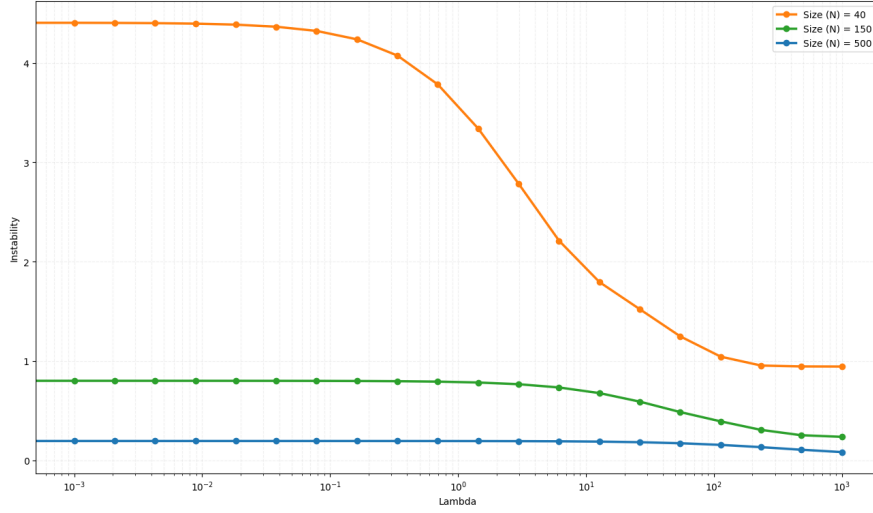


Figure 2: Effects of a larger dataset on instability.

While regularization is a powerful tool to enforce stability, statistical learning theory suggests that the intrinsic stability of an unregularized algorithm is heavily dependent on the sample size (N). To empirically test this dimension of the Law of Large Numbers within the context of algorithmic stability, the Leave-One-Out (LOO) experiment was repeated across three distinct dataset sizes: $N = 40$ (low-data regime), $N = 150$ (moderate-data regime), and $N = 500$ (high-data regime), while keeping the number of features constant ($D = 15$).

Low-Data Regime ($N = 40$): The unregularized model ($\lambda \approx 0$) is highly volatile, exhibiting an instability metric above 4.0. In this scenario, the removal of a single training point dramatically alters the decision boundary. Consequently, the model requires a significant amount of regularization ($\lambda > 10^1$) to be forced into a stable state.

Moderate-Data Regime ($N = 150$): The initial instability is substantially lower, and the regularization penalty gently guides the model toward stability.

High-Data Regime ($N = 500$): The algorithm demonstrates inherent stability almost entirely independent of the regularization strength. The blue curve remains virtually flat near zero across the entire logarithmic spectrum. Because the dataset is so heavily populated relative to the feature space ($N \gg D$), the removal of a single outlier through LOO validation is mathematically insufficient to meaningfully perturb the estimated OLS weights. These findings confirm that regularization primarily acts as a variance-reduction mechanism. Its utility is absolutely critical in low-data or high-dimensional regimes, but its stabilizing impact becomes increasingly marginal as the volume of training data grows sufficiently large to naturally suppress algorithmic sensitivity.

5 Comparative Analysis of L1 and L2 Regularization

To further explore how different penalty geometries influence both stability and model complexity, a comparative analysis between Ridge Regression (L2 penalty) and Lasso Regression (L1 penalty) was conducted. While both techniques aim to prevent overfitting by restricting the magnitude of the weight vector, their mathematical formulations induce fundamentally different behaviors in the hypothesis space. To visualize this, the models were evaluated on a higher-dimensional synthetic dataset ($N = 100, D = 40$), plotting both their generalization error and their induced sparsity (defined as the exact number of weights reduced to 0.0).

Generalization and the Bias-Variance Trade-off

The left panel of Figure 3 compares the Test Error of both models. Both algorithms successfully mitigate the initial overfitting, producing the characteristic "U-shape" curve. However, their trajectory-

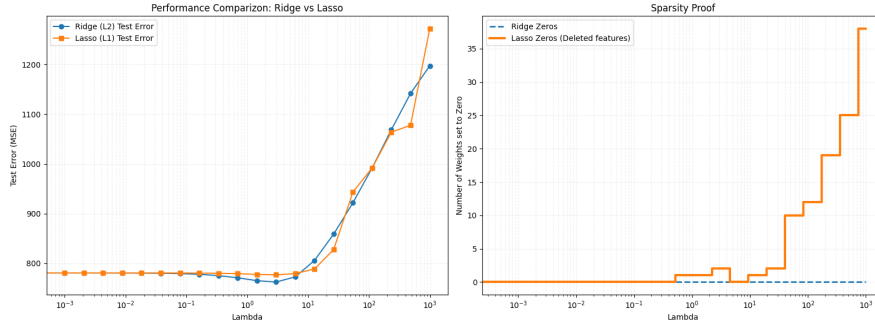


Figure 3: Performance comparison and sparsity proof

ries diverge slightly at higher regularization strengths. This behavior depends on the underlying true distribution of the data: if the true target function depends only on a small subset of features, Lasso tends to find a deeper minimum (better generalization) by aggressively ignoring the irrelevant noise variables.

The Geometry of Sparsity and Feature Selection

The most striking distinction between the two algorithms is demonstrated in the right panel of Figure 3. The L2 penalty of Ridge Regression shrinks the weights proportionally. Because the geometric constraint of the L2 norm is a hypersphere, the optimal solution rarely touches the axes of the parameter space. Consequently, the blue dashed line remains perfectly flat at zero: Ridge Regression never entirely eliminates a feature, regardless of how strong the λ penalty becomes. Conversely, the L1 penalty of Lasso Regression operates as a mathematically rigorous feature selection mechanism. Because the L1 constraint forms a geometric polytope with sharp corners (e.g., a diamond in 2D space), the optimization process—solved here via Coordinate Descent and Soft Thresholding—frequently finds its minimum exactly on the axes. As shown by the orange step-function, as λ increases, Lasso progressively “deletes” features, sequentially forcing their weights to absolute zero. This proves that while both L1 and L2 regularization successfully stabilize the algorithm to improve test error, L1 offers the distinct architectural advantage of producing sparse, highly interpretable models by actively filtering out irrelevant dimensions.

6 Real-World Application: The Diabetes Dataset

While synthetic datasets provide a pristine environment to isolate and demonstrate theoretical concepts, real-world data is inherently noisier, and the true underlying functions are rarely perfectly linear. To validate that the principles of algorithmic stability hold in practical scenarios, the base experiment was replicated using the Scikit-Learn Diabetes Dataset, a standard benchmark for regression tasks consisting of 442 samples and 10 baseline variables (features).

Observations on Real-World Data

The empirical results derived from the Diabetes dataset confirm the fundamental dynamics observed in the synthetic experiments, though the curves exhibit the characteristic subtlety of real-world data:

1. **Initial Instability:** At $\lambda = 0$ (OLS), the model exhibits an initial instability metric of roughly 0.307 and a test error of 3368. While not as dramatically overfit as the synthetic low-data regime, the unregularized model still leaves room for improvement.
2. **Variance Reduction:** As λ increases toward the range of 10^1 , the instability metric steadily decreases. This confirms that even on real-world data, the L2 penalty successfully forces the algorithm to become less sensitive to the removal of individual patients from the training set.
3. **The Minimum Test Error:** The reduction in instability directly correlates with improved generalization. The test error reaches its minimum (approximately 3316) at $\lambda \approx 25.9$. This

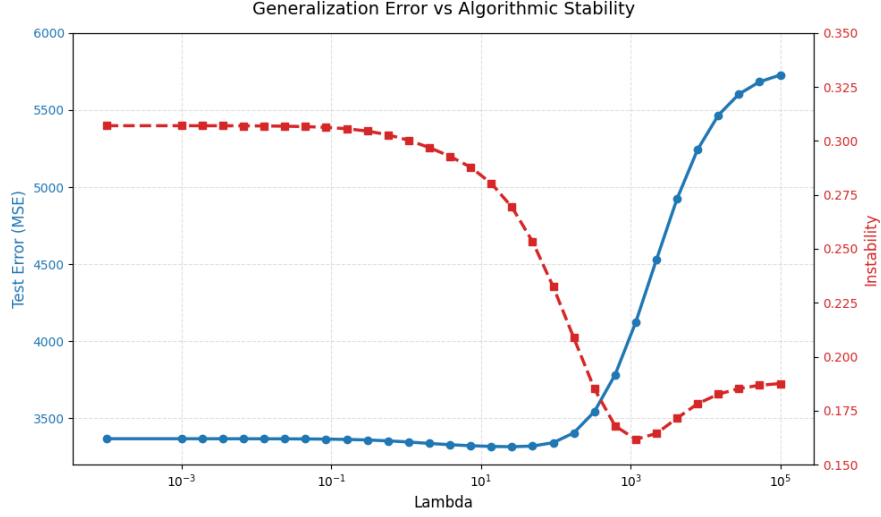


Figure 4: Impact of penalty, in a real world dataset

is the optimal point where the model has been stabilized just enough to ignore dataset noise without sacrificing its ability to map the features to the disease progression.

4. **The Onset of Bias:** Pushing λ beyond 10^2 causes the test error to climb steeply, eventually plateauing near 5727. The instability also reaches a floor, indicating that the model has become excessively rigid and is now underfitting the complex physiological data.

This experiment successfully bridges the gap between theory and practice, demonstrating that the framework of algorithmic stability is not merely a mathematical abstraction, but a robust operational principle for tuning models on real-world datasets.

7 Conclusions

This empirical study investigated the profound connection between generalization performance and algorithmic stability, as theorized by Bousquet and Elisseeff. By tracking the behavior of unregularized OLS and Ridge Regression across varying penalty strengths, the experiments consistently demonstrated that regularization acts primarily as a stabilizing mechanism. The core finding is that an algorithm’s ability to generalize to unseen data is intrinsically linked to its insensitivity to minor perturbations in the training set. However, stability alone is not the goal; absolute stability results in underfitting. The optimal model is always found at the critical intersection where regularization suppresses algorithmic variance just enough to prevent overfitting, without introducing excessive bias. Furthermore, the extensions of this study confirmed that the necessity of explicit regularization decreases as the dataset size grows (due to the natural stabilizing effect of large samples) and highlighted the unique architectural advantage of L1 regularization in enforcing sparsity and feature selection. Ultimately, viewing generalization through the lens of algorithmic stability provides a highly operational and intuitive framework for designing and optimizing robust machine learning models.

References

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